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Triphenylene-Involved π-Extension Combining With Phenyl-Blocking Enhances the Stability of MR-TADF Emitter: Top-Emitting OLED Realizes EQE Approaching 60% With BT.2020 Green Gamut and Long Lifetime

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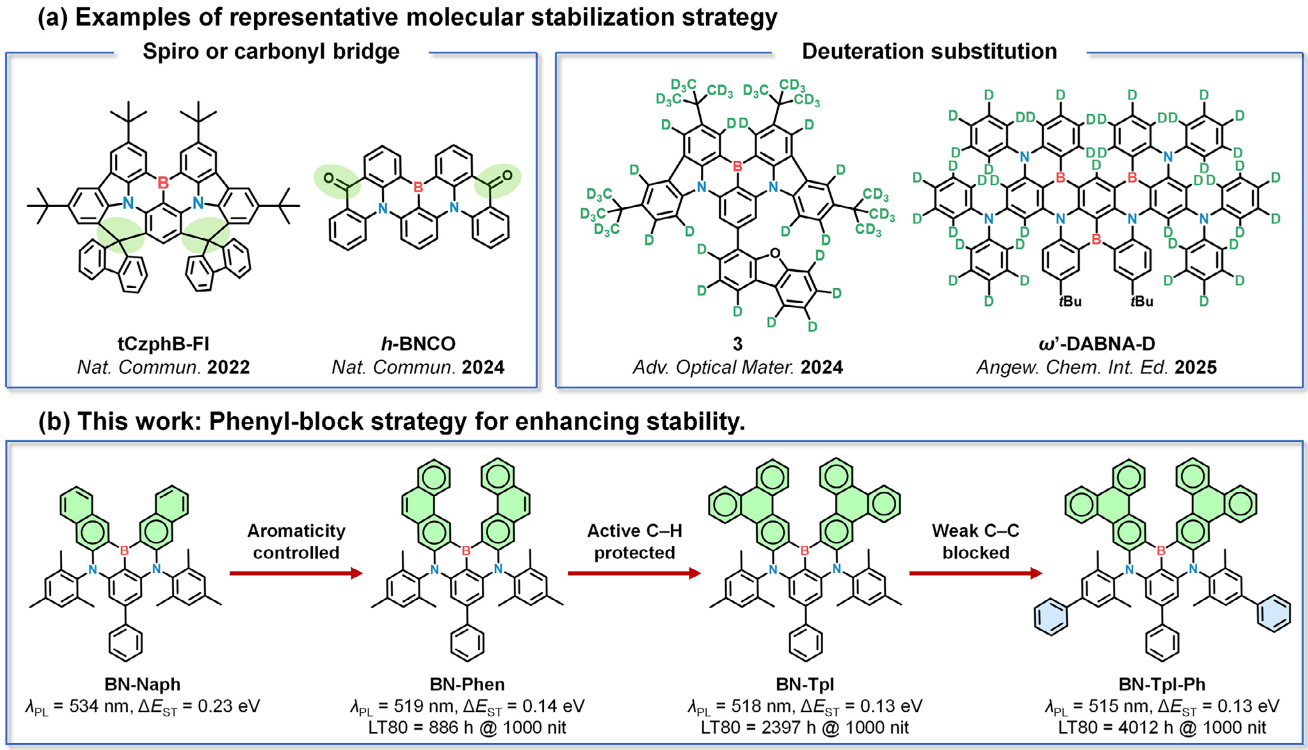
ABSTRACT

The realization of ultrahigh definition displays necessitates pure-green organic light-emitting diodes (OLEDs) with simultaneously exceptional efficiency, high color purity, and long-term operational stability. Herein, we develop a series of pure-green multi-resonance thermally activated delayed fluorescence (MR-TADF) emitters based on the boron/nitrogen-embedded polycyclic aromatic hydrocarbon (BN-PAH), designed through a rational moderate π-extension and peripheral phenyl blocking strategy. The molecular designs afford narrowband pure-green emission with a full-width at half-maximum (FWHM) of nearly 20 nm and high photoluminescence quantum yields (Φ_{PL}s, up to 95%). By systematically blocking the redox-active positions with phenyl groups, the emitters exhibit significantly enhanced electrochemical and photochemical stability. In bottom-emitting OLEDs, the optimized emitter BN-Tpl-Ph achieves a maximum external quantum efficiency (EQE$_{max}$) of 33.8% and long operational lifetime (LT80 = 4012 h at 1000 cd m^{-2}). Notably, in a top-emitting OLED configuration, BN-Tpl-Ph delivers a pure-green emission with a Commission Internationale de l'Eclairage (CIE) y-coordinate of 0.78, a high EQE$_{max}$ up to 59.2%, and an LT80 of 409 h at 5000 cd m^{-2}. This work reveals the effectiveness of molecular design strategies that combine moderate π-extension with peripheral phenyl blocking for developing high-performance pure-green MR-TADF emitters.

1 | Introduction

In order to meet the stringent color requirements of the BT.2020 standard for ultrahigh definition (UHD) organic light-emitting diode (OLED) displays, the development of emitters simultaneously exhibiting high efficiency, wide color gamut, and exceptional stability remains a critical imperative [1–3]. Since the pioneering work by Hatakeyama et al. in 2016, multi-resonance (MR) thermally activated delayed fluorescence (TADF) emitters have garnered considerable interest due to their suppressed molecular relaxation and consequently narrow full width at half maximum (FWHM) [4]. This unique photophysical behavior originates from the atomically alternative separation of the highest occupied molecular orbital (HOMO) and the lowest unoccupied molecular orbital (LUMO), which minimizes the bonding/antibonding character, suppresses vibrational coupling and relaxation, and consequently affords narrow FWHM [5, 6].

While remarkable progress has been achieved in blue MR materials, the development of pure-green MR-TADF emitters remains



SCHEME 1 | Molecular design concept.

challenging due to their strict requirement toward both emission peak position and spectral shape [3]. Derived from the typical blue or sky-blue MR frameworks such as DABNA-1, ν-DABNA and BCz-BN [4, 7, 8], various molecular engineering strategies have been developed for realizing bathochromically shifted emission, such as peripheral modification [9–16], π-extension [17–20] and locking strategy [21, 22]. Among these, π-extension offers a particular straightforward approach, with purely hydrocarbon aromatic skeletons being attractive owing to their synthetic accessibility. By rationally modulating the number of Clar sextets, several green emitters have been reported [23–27]. For example, Wang et al. introduced a pyrene unit into the BCz-BN scaffold to construct a circular polarization MR emitter BN-Py, which exhibited narrowband green emission peaking at 527 nm with FWHM of 35 nm [26]. Recently, our group reported a green emitter DBN-NaPh-d based on a double-boron skeleton incorporating naphthalene moieties, which realized narrowband pure-green emission with CIE coordinates approaching the BT.2020 standard [19]. Despite these advances, excessive π-extension often lowers the triplet energy (T_1), leading to an enlarged singlet-triplet energy gap (ΔE_{ST}) and weakened TADF characteristic, thereby impeding triplet exciton recycling.

Apart from spectral challenges, another major obstacle for MR emitters is the presence of reactive sites within the boron/nitrogen-doped polycyclic aromatic hydrocarbon (BN-PAH) skeleton, which compromises device operational durability. To enhance chemical stability for pure-organic emitters, several molecular design methodologies have been developed. One reliable molecular decorating strategy involves locking chemical reactive sites via spiro-carbon bridges or carbonyl groups to suppress the excited-state distortion and the vibration modes caused by intramolecular charge transfer (ICT) effect (Scheme 1). For instance, Zhang et al. employed spiro-carbon modification to block the peripheral phenyl rings, and the multi-locked emitter tCzphB-Fl achieved significantly improved device stability [21]. Recently, Zhang et al. incorporated carbonyl groups into the conventional BNCZ framework to modulate molecular rigidity and excited-state property, producing an emitter *h*-BNCO with a significantly red-shifted emission along with a 170-fold improvement in device lifetime compared to the control [22]. Another approach is to use deuteration modification, which would decrease the vibrational zero-point energy (ZPE) and suppress degradation pathways [28–31]. For example, Tang et al. reported a green mono-boron MR-TADF emitter, and the deuterated analogue exhibited a sevenfold enhancement in device lifetime [30]. Very recently, our group adopted a sequential borylation strategy to develop a tri-boron emitter ω'-DABNA-D, achieving a 1.7-fold lifetime improvement over its nondeuterated counterpart [31]. However, these strategies usually involve complex synthetic routes and harsh conditions.

In this work, we attempt to develop a series of pure-green MR-TADF emitters with a rational π-extension and a peripheral phenyl blocked strategy. Specifically, the incorporation of naphthalene units into the DABNA skeleton resulted in an excessively red-shifted emitter with diminished TADF character. While adopting a moderate aromatic sextet, the emitters fused with phenanthrene and triphenylene not only showcase appropriate emission wavelengths approaching the BT.2020 standard, but also maintain favorable TADF properties. To further shield the BN-PAH framework against redox reactants, a phenyl blocked strategy was introduced to enhance operational stability without altering the intrinsic electronic structure and sacrificing color-purity. The proof-of-concept emitter BN-Tpl-Ph demonstrated markedly improved device lifetime, with ternary OLEDs achieving an LT80 (time to 80% of the initial luminance) of 240 h at an initial luminance of 5000 cd m^{-2}. Moreover, the top-emitting OLED delivered ultra-pure green emission (CIEy = 0.78), high current efficiency (CE) of 218.0 cd A^{-1}, and exceptional operational stability with LT80 up to 409 h at an initial luminance of 5000 cd m^{-2}.

2 | Results and Discussion

Derived from the mono-boron MR-TADF skeleton DABNA, the emitters BN-Naph, BN-Phen, BN-Tpl, and BN-Tpl-Ph were facilely prepared via Buchwald-Hartwig C–N coupling reactions, followed by lithium-free borylation in decent yield. The detailed synthetic procedures and characterization data are provided in the Supporting Information (Schemes S1–S4, Figures S16–S36). The high-performance liquid chromatography (HPLC) measurement was conducted to further verify the compound purity (Figures S37–S39). The crystal structures and molecular packing motifs of the emitters are depicted in Figure S1 and Table S1 [32]. These compounds display rigid and planar BN-PAH frameworks, which effectively suppress molecular twisting and vibrational coupling, thereby enhancing structural stability. Moreover, the nearly orthogonal orientation of the peripheral phenyl rings relative to the BN-PAH plane minimizes π-π overlap, as evidenced by large π-π distances (3.60–3.87 Å) and displaced angles (70.8–74.2°). This loose packing is expected to reduce aggregation-caused quenching (ACQ) and suppress triplet exciton loss in the solid state.

To elucidate the influence of the aromatic segment on photophysical behavior, we first investigated the corresponding hydrocarbon building blocks. As shown in Figure 1a, despite containing more fused benzene rings, phenanthrene and triphenylene skeletons showcased higher triplet (T_1) energies than naphthalene (T_1 = 2.62 eV, 2.69 eV, and 2.90 eV for naphthalene, phenanthrene, and triphenylene, respectively) [33–35]. The increased T_1 energy states can be attributed to their nonlinear geometries, which restrict excessive π-electron delocalization. The Clar structures depicted in the right panel of Figure 1a illustrate the distribution of aromatic π-sextets, which possess six π-electrons localized in a single benzene-like ring separated from adjacent rings by formal C–C single bonds [36–38]. As presented in naphthalene, the migrating π-sextet between two rings signifies the adequate π-electron delocalization, thus the two rings are totally equivalent and result in an identical nucleus-independent chemical shift (NICS) value of –10.23 ppm. In contrast, the central ring (Ring B) in phenanthrene and triphenylene segments differs distinctly from the outer rings (Ring A). For phenanthrene, Ring B appears as a six-membered ring containing a localized double bond, whereas in triphenylene, it is essentially an "empty ring". Accordingly, the NICS(0) values decrease markedly to –7.11 ppm for phenanthrene and further to only –2.69 ppm for triphenylene, indicating substantially reduced aromatic character in these central rings. These observations suggest that the central rings in phenanthrene and triphenylene inhibit excessive π-electron delocalization, thereby mitigating the undesired energy gap narrowing and ΔE_{ST} increasing.

To further assess the electronic impact of these conjugation fragments within the complete molecular structures, density functional theory (DFT) and time-dependent DFT (TD-DFT) calculations were performed at the B3LYP/def2svp level for BN-Naph, BN-Phen, BN-Tpl, and BN-Tpl-Ph (Figure 1b). These emitters exhibit large oscillator strengths (f, 0.2020 for BN-Naph; 0.2173 for BN-Phen; 0.2899 for BN-Tpl; 0.3255 for BN-Tpl-Ph), indicative of the efficient $S_0 \rightarrow S_1$ transitions and suggesting potentially high photoluminescence quantum yields (Φ_{PL}s). The frontier molecular orbital (FMO) distributions display a characteristic alternating pattern on the BN-PAH skeleton, confirming their MR nature. Particularly, BN-Naph presents more extended delocalization of the FMOs (in dashed circle) compared to the others (i.e., Ring B in the isolated fragment). Consequently, BN-Naph possesses a noticeably narrower energy gap (E_g) and a larger ΔE_{ST} (0.46 eV) than the others (ΔE_{ST} = 0.38 eV for BN-Phen as well as 0.37 eV for both BN-Tpl and BN-Tpl-Ph, respectively), implying an excessive bathochromic shift and inferior TADF characteristics. Furthermore, in the triphenylene-fused systems, the HOMO and LUMO are primarily localized on different outer rings (Ring A), a delocalization feature that may be conducive to enhancing molecular stability [39]. The further natural transition orbital (NTO) analysis showed that the S_1 and T_1 are dominated by HOMO-LUMO transitions in BN-Phen, BN-Tpl, and BN-Tpl-Ph, with a high contribution of over 90% (Figure S2). It is noticed that BN-Naph possesses a reduced HOMO-LUMO contribution in T_1 (82.7%) compared to that of S_1 (98.6%), indicating an admixture of local-excitation character. This change in electronic configuration lowers the T_1 energy relative to S_1 and thus increases ΔE_{ST} for BN-Naph.

Given the balance of optical and TADF properties, we therefore investigated the vibrational coupling and structural rigidity of BN-Phen, BN-Tpl, and BN-Tpl-Ph. The Huang-Rhys factor (S) of the vibrational modes and reorganization energy (λ) values were calculated and analyzed (Figure 1c). The vibrational coupling for these emitters is concentrated in the low-frequency region (less than 500 cm^{-1}), corresponding to the skeleton twisting vibration of the BN-PAH backbone. Whilst a distinct difference emerges in the high-frequency region (over 3000 cm^{-1}), associated with C–H scissoring/stretching vibrations. The triphenylene-based derivatives BN-Tpl and BN-Tpl-Ph exhibit substantially reduced S factors in this region, corroborated by lower total reorganization energies (λ_{total} = 0.18 eV for BN-Phen vs. 0.16 eV for both BN-Tpl and BN-Tpl-Ph). These results indicate that the extended, disk-shaped triphenylene moiety effectively blocks the active 9,10-positions of the phenanthrene unit, imposing better structural constraint and reducing nonradiative vibrational dissipation. Such enhanced rigidity directly contributes to both the narrower emission bandwidths and the improved stability.

Additionally, to further evaluate structural robustness, bond dissociation energies (BDEs) for critical redox active positions, i.e., C–N and C–C bonds at the *para*-positions relative to the boron atom, were computed (Figure 1d). While the BDEs of the C–N bonds remain nearly constant (~83 kcal/mol) across the series, the C–C bonds exhibit significant variation. The BDE for the C–C bond in BN-Tpl-Ph was calculated to be 114.5 kcal/mol, substantially higher than those in BN-Phen (98.8 kcal/mol) and BN-Tpl (99.5 kcal/mol). This marked increase highlights the stabilizing effect of the peripheral phenyl substituent in BN-Tpl-Ph, which provides steric shielding and reinforces the bond at the redox-active site.

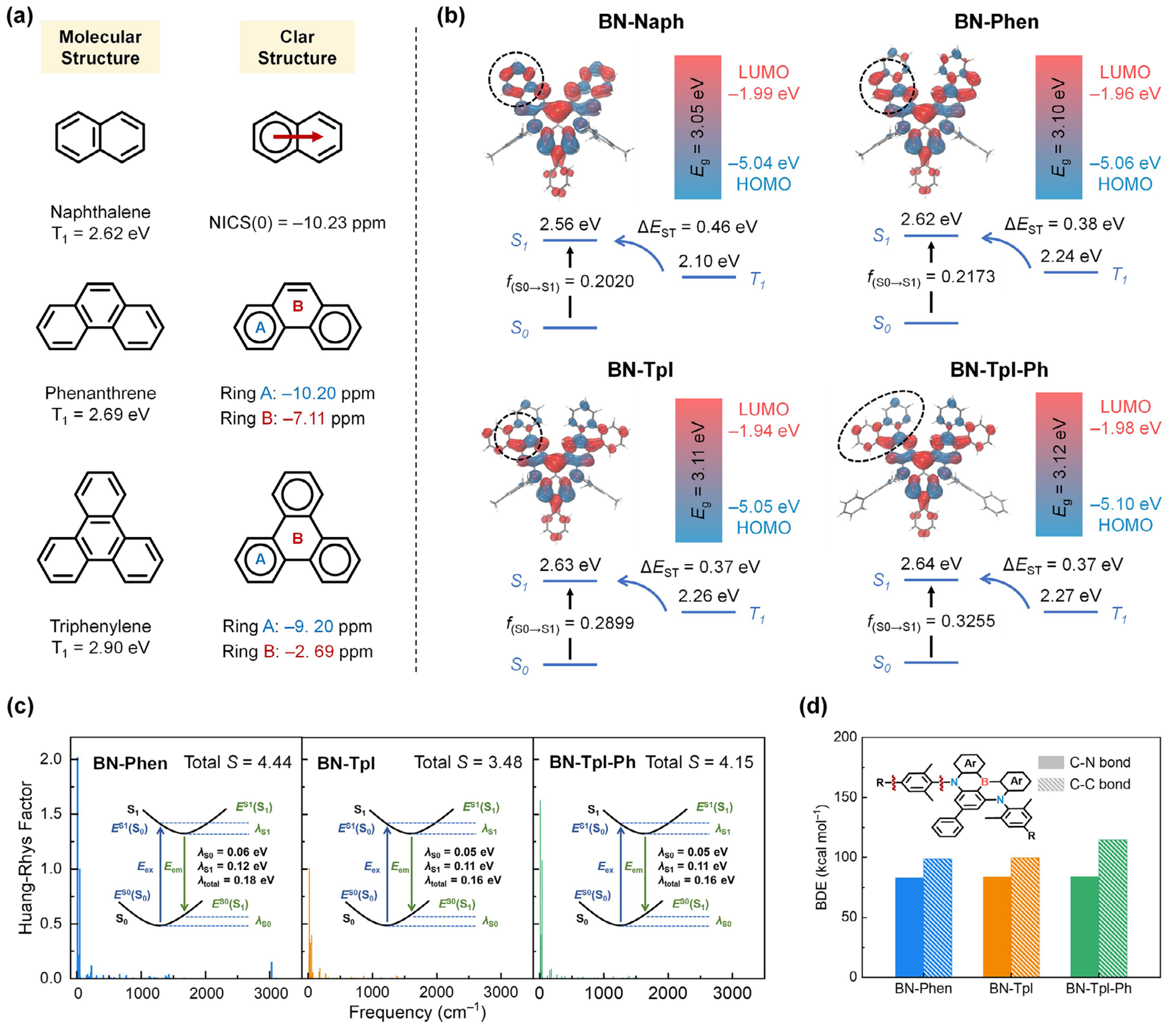


FIGURE 1 | (a) Molecular structures, Clar structures of naphthalene, phenanthrene, and triphenylene units, and their T_1 energy states, NICS(0) values. (b) Calculated HOMO-LUMO distribution with energy level gap (E_g) values and oscillator strength (f), excited-state energies of BN-Naph, BN-Phen, BN-Tpl, and BN-Tpl-Ph. (c) Huang-Rhys factor versus frequency plots and reorganization energies for the $S_1 \rightarrow S_0$ transition. (d) Bond dissociation energy of C–N and C–C bonds.

The photophysical properties of the emitters were investigated in dilute toluene solution at a concentration of 1.0×10^{-5} M. The UV/Vis spectra revealed sharp and intense absorption bands with the peak wavelengths at 501 nm, 500 nm, and 499 nm for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively (Figure 2a–c; Table 1). Extending the π-system within the DABNA framework led to a red-shifted emission, yielding narrowband green luminescence for the new compounds. Linear extension via the naphthalene moiety significantly expanded the conjugation length, resulting in an excessively red-shifted emission (λ_{em} = 534 nm) for BN-Naph. Concurrently, the T_1 energy was markedly reduced, giving rise to an enlarged ΔE_{ST} of 0.23 eV and a weakened TADF character (Figure S3). On the contrary, the angular structure of phenanthrene and the disc-shaped architecture of triphenylene effectively limited excessive π-delocalization, thereby affording a moderate red-shifted emission while preserving the TADF feature. Accordingly, the fluorescence spectra in toluene revealed pure-green emissions peaking at 519 nm, 518 nm, and 515 nm for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively, accompanied by narrow FWHMs of 22 nm, 20 nm, and 20 nm, and small ΔE_{ST} values of 0.14 eV, 0.13 eV, and 0.13 eV (Figure S4). The emitters demonstrated minimal variations of the emission peaks in solvents with different polarities, manifesting their dominant short-range charge transfer (SRCT) characteristics (Figure S5, Table S2). In low-polarity solvents, their spectra feature well-resolved characteristics, composed of a sharp 0–0 transition main peak and a 0–1 transition shoulder peak. As high-polarity solutions intensify vibrational relaxations, the FWHMs are broadened, thereby including the shoulder peaks. As a result, the shoulder peaks become insignificant in high-polarity solvents.

In order to shed light on the stability discrepancy of these emitters, further electrochemical and photochemical stability was experimentally probed [40]. A spectro-electrochemical stability

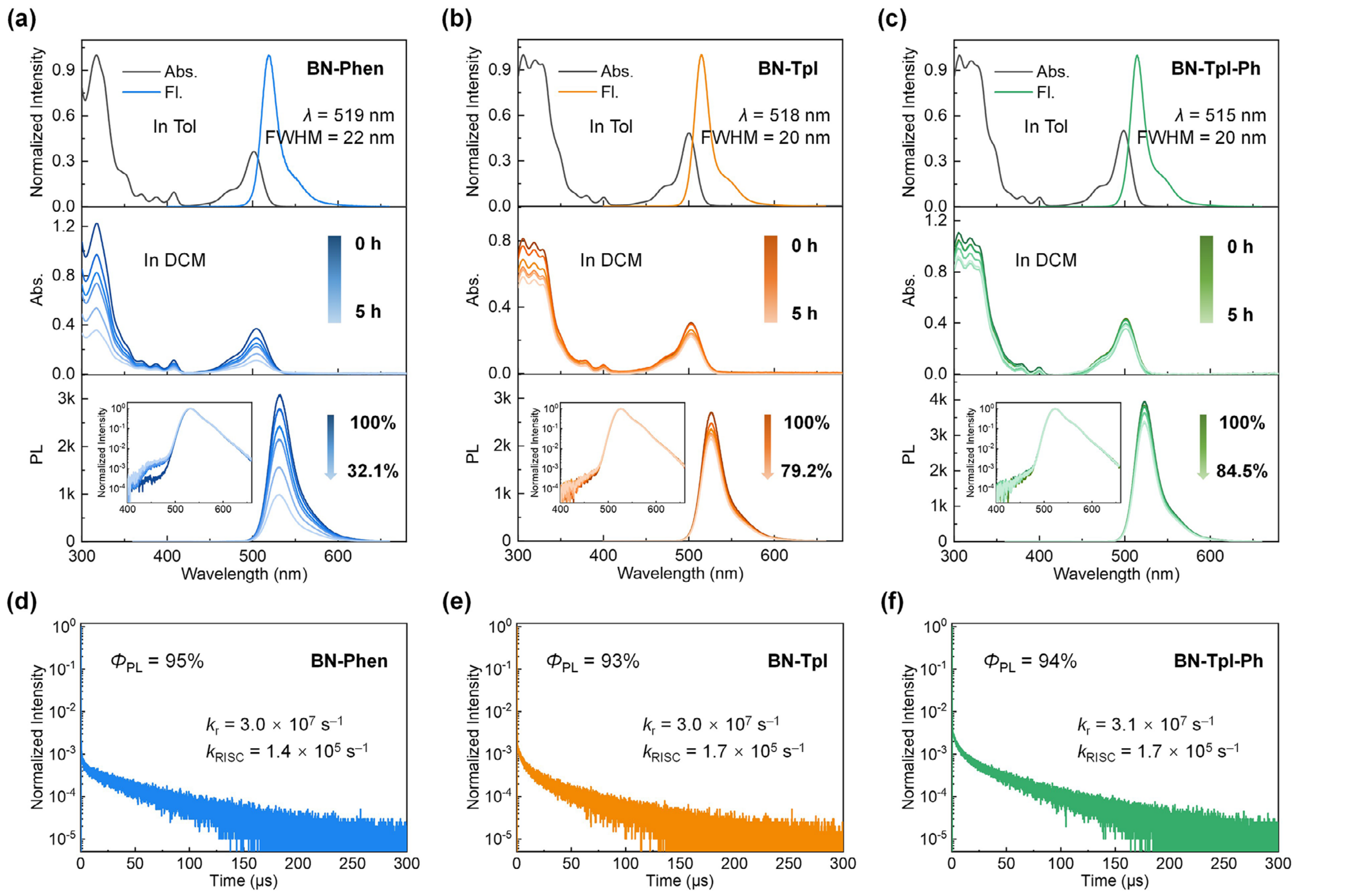


FIGURE 2 | Photophysical properties. Absorption (Abs.: 298 K) and fluorescence (Fl.: 298 K) spectra in toluene solution at 1.0×10^{-5} M; absorption and PL spectra of 1.0×10^{-4} M (N_2-saturated DCM containing 0.10 M tetrabutylammonium hexafluorophosphate) obtained during oxidative electrolysis (0–5 h) at an anodic potential of 2.0 V for (a) BN-Phen, (b) BN-Tpl and (c) BN-Tpl-Ph (Inset: the PL spectra with logarithmic coordinates after normalization). Transient PL spectra (298 K) of 3 wt% emitter:DMIC-TRZ films for (d) BN-Phen, (e) BN-Tpl, and (f) BN-Tpl-Ph.

TABLE 1 | Summary of photophysical data.

Compound	λ_{abs}[a] (nm)	λ_{em}[a] (nm)	FWHM[a] (nm)	ΔE_{ST}[b] (eV)	Φ_P[c] (%)	τ_p[c] (ns)	τ_d[c] (μs)	k_r[c] ($\times 10^7$ s^{-1})	k_{RISC}[c] ($\times 10^5$ s^{-1})
BN-Phen	501	519	22	0.14	95	18.5	33.4	3.0	1.4
BN-Tpl	500	518	20	0.13	93	12.0	32.0	3.0	1.7
BN-Tpl-Ph	499	515	20	0.13	94	14.2	35.6	3.1	1.7

[a]Measured in toluene solution at 1.0×10^{-5} M (298 K).
[b]Determined from the onsets of the fluorescence (77 K) and phosphorescence spectra (77 K) in dilute toluene (1.0×10^{-5} M).
[c]Measured in doped films (3 wt.% in DMIC-TRZ).

measurement was conducted by applying an anodic potential of 2.0 V to a nitrogen-saturated dichloromethane (DCM) solution containing 10^{-4} M of the corresponding emitters and 0.1 M tetrabutylammonium hexafluorophosphate as supporting electrolyte (Figure 2a–c). Over 5 h timespan, the absorption and fluorescence intensities decreased with clear trends in signal attenuation, i.e., BN-Phen retained only 32.1% of its initial intensity, while BN-Tpl and BN-Tpl-Ph retained 79.2% and 84.5%, respectively. Notably, the emergence of a new emission band at around 450 nm for BN-Phen suggests oxidative degradation specifically at the phenanthrene moiety, a process effectively mitigated in the triphenylene-based analogues. This trend was corroborated by controlled photostability measurements (Figure S6) and aligns well with the theoretical simulation.

Solid-state photophysical behaviors were investigated by dispersing the emitters into a host matrix of 5-(3-(4,6-diphenyl-1,3,5-triazin-2-yl)phenyl)-7,7-dimethyl-5,7-dihydroindeno[2,1-*b*]carbazole (DMIC-TRZ) with an optimized doping concentration of 3 wt% (Figure S7 and Table S3). The fluorescence spectra of these films exhibited a modest red-shift and slight broadening relative to solution data. High Φ_{PL}

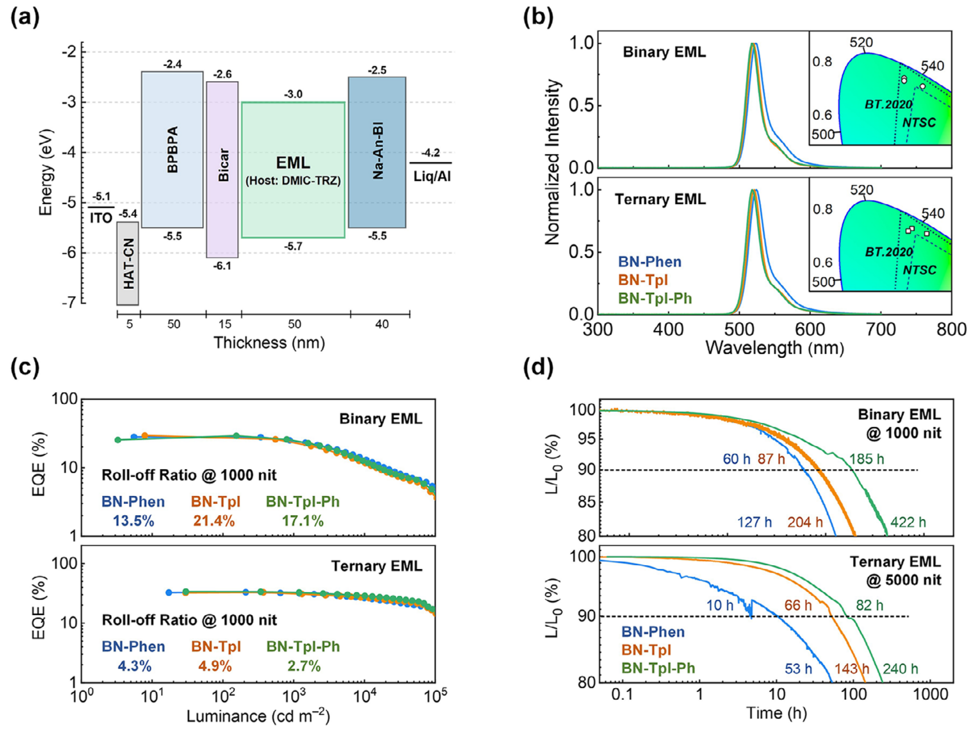


FIGURE 3 | BE-OLED performances of BN-Phen, BN-Tpl, and BN-Tpl-Ph. (a) Device structures and energy level diagrams. (b) Electroluminescence spectrum at 1000 cd m^{-2} with color coordinates. (c) EQE-luminance of the OLEDs with the roll-off ratios at 1000 cd m^{-2}, calculated by (EQE_{max}—EQE_{1000})/EQE_{max}. (d) Luminance-deterioration curves of the OLEDs based on binary EML systems (L_0 = 1000 cd m^{-2}) and ternary EML systems (L_0 = 5000 cd m^{-2}).

values were measured to be 95%, 93%, and 94% for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively, establishing a robust foundation for fabricating high-efficiency pure-green OLEDs. The transient photoluminescence (TRPL) decay profiles of all emitters followed similar biexponential kinetics. To gain deeper insight into the TADF behavior, key photophysical rate constants were extrapolated (Figure 2d–f). Consistent with the theoretical simulation, BN-Naph reveals poor TADF property with rather slow k_{RISC} of only 10^3 s^{-1} (Figure S3). In contrast, BN-Phen, BN-Tpl, and BN-Tpl-Ph exhibited much better while comparable exciton dynamics, with fast radiative decay rates (k_r) on the order of 10^7 s^{-1} and reverse intersystem crossing rates (k_{RISC}) of 10^5 s^{-1}. The excellent photophysical performance underscores the strong potential of these emitters for OLED applications.

Prior to device fabrication, the electrochemical properties of BN-Phen, BN-Tpl, and BN-Tpl-Ph were evaluated by cyclic voltammetry (CV) in anhydrous DCM (Figure S8). The HOMO energy levels were determined to be –5.18, –5.18, and –5.21 eV for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively. Corresponding LUMO levels, derived from the HOMO values and the optical E_g, were calculated as –2.80, –2.80, and –2.81 eV (Table S4). The closely aligned frontier orbital energies among the three emitters suggest similar charge-injection and transport characteristics in devices. All emitters demonstrated excellent electrochemical stability over 100 cyclic voltammetry cycles (Figure S9) and exhibited high thermal decomposition temperatures ($T_d > 400$°C, Figure S10, Table S4), indicating outstanding electrochemical and thermal stabilities.

To evaluate electroluminescent (EL) performances, bottom-emitting OLEDs (BE-OLEDs) were fabricated with the optimized device structure of ITO/HAT-CN (5 nm)/BPBPA (50 nm)/Bicar (15 nm)/emitting layer (EML, 50 nm)/Na-AN-BI (40 nm)/Liq (2 nm)/Al. Here, HAT-CN and Liq function as the hole-injection layer (HIL) and electron-injection layer (EIL), respectively. BPBPA and Bicar were applied as the hole-transporting layer (HTL), and Na-AN-BI served as the electron-transport layer (ETL). The corresponding energy-level diagrams are depicted in Figure 3a, and the molecular structures involved in OLEDs are shown in Figure S11. Two EML configurations were studied, composed of the emitter:DMIC-TRZ = 3:97 for binary emitting systems, while emitter:Ir(ppy)$_3$:DMIC-TRZ = 3:10:87 for ternary emitting systems, respectively. Key OLED performance parameters are summarized in Table 2.

The EL spectra of all devices closely resemble their PL profiles, exhibiting narrowband pure-green emission with peaks between 517–524 nm and FWHMs of 22–26 nm (Figure 3b). Notably, the binary device incorporating BN-Tpl achieved the Commission Internationale de l'Eclairage (CIE) coordinates of (0.18, 0.74), approaching the BT.2020 green standard (0.17, 0.79). All emitters delivered similar device efficiency trends and roll-off behaviors in both binary and ternary configurations (Figure 3c and Figure S12). For binary EMLs, the maximum external quantum efficiencies (EQE_{max}) reached 28.1% for BN-Phen, 29.4% for BN-Tpl, and 29.2% for BN-Tpl-Ph. Ternary EMLs, employing Ir(ppy)$_3$ as sensitizer, further boosted EQE_{max} to 32.9%, 32.9%, and 33.8%, respectively, while substantially mitigating efficiency roll-off. At a practical

TABLE 2 | Summary of OLED performances.

Device structure	Emitter	V_{on}[a] (V)	λ_{EL}[b] (nm)	FWHM[b] (nm)	CIE[b] (x, y)	$EQE_{max/1000/10000}$ (%)	PE_{max} (lm W^{-1})	CE_{max} (cd A^{-1})	LT_{80}[c] (h)
Binary EML	BN-Phen	2.4	523	26	(0.23, 0.71)	28.1/24.3/13.4	143.9	109.9	127
	BN-Tpl	2.4	520	22	(0.18, 0.74)	29.4/23.1/11.6	144.9	110.7	204
	BN-Tpl-Ph	2.4	517	22	(0.18, 0.73)	29.2/24.2/12.1	122.4	107.8	422
Ternary EML	BN-Phen	2.4	524	26	(0.24, 0.71)	32.9/31.5/26.1	166.8	128.8	53
	BN-Tpl	2.4	520	23	(0.20, 0.73)	32.9/31.3/26.7	157.3	122.6	143
	BN-Tpl-Ph	2.4	518	23	(0.19, 0.72)	33.8/32.9/28.5	161.1	123.1	240
TE-OLED	BN-Tpl-Ph	2.4	520	19	(0.12, 0.78)	59.2/59.0/50.6	285.3	218.0	409

[a]Turn-on voltage at 1 cd m^{-2}.
[b]Electroluminescence peak wavelength, full-width at half-maximum, and CIE coordination at 1000 cd m^{-2}.
[c]Operational lifetimes at the initial luminance of 1000 cd m^{-2} for the binary EML system and 5000 cd m^{-2} for both the ternary EML system and TE-OLED.

brightness of 1000 cd m^{-2}, the ternary devices retained high EQEs of 31.5%, 31.3%, and 32.9% for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively, underscoring the effectiveness of the sensitized system in suppressing triplet-related losses.

The Förster resonance energy transfer (FRET) processes were evaluated to elucidate the exciton kinetics within the ternary EML systems. As shown in Figure S13, significant spectral overlap between the PL spectrum of $Ir(ppy)_3$ sensitizer and the absorption spectra of these MR emitters enabled a large Förster radius (R_0), leading to highly efficient energy transfer in the ternary-emitting systems (Figure S14 and Table S5). The doped films maintained high Φ_{PL} values of 93%, 93%, and 97% for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively. TRPL spectra revealed rapid FRET rates (k_{FRET}) on the order of 10^6 s^{-1}, with donor-acceptor distances well within R_0, ensuring near-unity exciton utilization in all three emitting systems [41, 42].

Operational stabilities were assessed through lifetime measurements under constant current driving (Figure 3d). Despite the same molecular motif, comparable exciton kinetics, and EL efficiencies, the three emitters exhibited pronounced differences in device durability. For binary OLEDs measured at an initial luminance of 1000 cd m^{-2}, the LT80 (time to 80% of the initial luminance) values were measured to be 127 h for BN-Phen and 204 h for BN-Tpl, while further prolonged to be 422 h for BN-Tpl-Ph. A similar tendency was observed in the ternary devices at an initial luminance of 5000 cd m^{-2}, with LT80 values of 53 h, 143 h, and 240 h for BN-Phen, BN-Tpl, and BN-Tpl-Ph, respectively. Applying the degradation acceleration model relating luminance to lifetime and assuming an acceleration factor n = 1.75, the LT80 of BN-Tpl-Ph-based ternary device was estimated to be 4012 h at an initial luminance of 1000 cd m^{-2}, confirming the effectiveness of the molecular design strategy and highlighting its superior stability for potential practical applications [43–49].

To further demonstrate the capability of these pure-green emitters in high-resolution displays, a top-emitting OLED (TE-OLED) using BN-Tpl-Ph as emitter was fabricated with a tailored microcavity structure: ITO/Ag/ITO (40 nm/140 nm/15 nm)/HAT-CN (5 nm)/BPBPA (135 nm)/Bicar (15 nm)/EML (emitter:$Ir(ppy)_3$:DMIC-TRZ = 3:10:87, 35 nm)/Na-AN-BI:Liq (50:50, 45 nm)/Yb (3 nm)/Ag (12 nm)/BPBPA (75 nm) (Figure S15). Benefiting from the effective vibrational suppression within the microcavity structure, the FWHM of the optimized TE-OLED was reduced to only 19 nm, elevating the CIEy coordinate to 0.78 (Figure 4a). The TE-OLED exhibited excellent spectral stability over a wide voltage range across 2–8 V (Figure 4b), indicating balanced charge transport over the device structures and efficient energy transfer within the EML. The device delivered extremely high efficiency, with a maximum current efficiency (CE_{max}) of up to 218.0 cd A^{-1}, EQE_{max} of 59.2%, and maximum power efficiency (PE_{max}) of 285.3 lm W^{-1}, without any external optical extraction structure (Figure 4d,e, Table 2). Importantly, at the initial luminance of 5000 cd m^{-2}, the LT90 lifetime of the TE-OLED persisted for 202 h and LT80 for 409 h, confirming the stability of BN-Tpl-Ph (Figure 4f). The high device efficiency and very long lifetime of BN-Tpl-Ph were attributed to its balanced carrier transport ability and sufficient exciton utility, and the intrinsic narrow spectrum enabled high photon utilization efficiency in the TE-OLED, suppressing the exciton annihilation and aging processes. This work underscores the effectiveness of molecular design strategies that integrate extended, rigid PAH frameworks with judicious peripheral substitution for developing durable, high-performance OLED emitters.

3 | Conclusion

To conclude, the combination of moderate π- extension with triphenylene units and peripheral phenyl blocking provides an effective strategy for simultaneously achieving narrowband pure-green emission and high operational stability in MR-TADF materials. The optimized emitter BN-Tpl-Ph exhibited excellent photophysical properties, suppressed degradation pathways, and superior device performance. Notably, TE-OLED based on BN-Tpl-Ph achieves near-BT.2020 color purity (CIEy = 0.78), high efficiency (CE_{max} = 218.0 cd A^{-1}, EQE_{max} = 59.2%), and prolonged operational lifetime (LT80 = 409 h at an initial luminance of 5000 cd m^{-2}), highlighting its promise for next-generation OLED displays. This work provides a feasible strategy for developing durable and high-performance MR-TADF emitters.

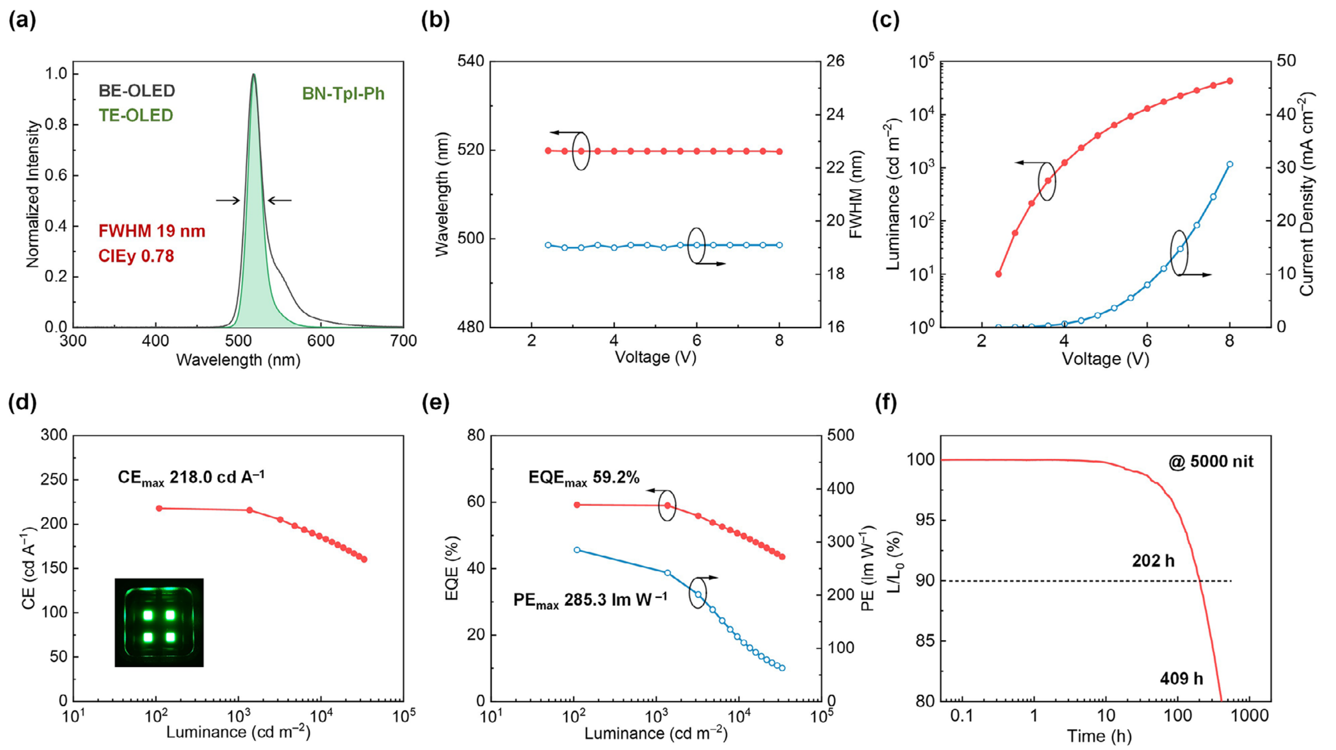


FIGURE 4 | TE-OLED performances of BN-Tpl-Ph. (a) EL spectra of BE-OLED and TE-OLED. (b) Wavelength-voltage-FWHM curves among 2–8 V. (c) Luminance-voltage-current density curves. (d) Current efficiency-luminance and (e) EQE-luminance-PE curves. (f) Luminance-deterioration curves ($L_0 = 5000$ cd m^{-2}).

Author Contributions

Jiahui Liu: writing – original draft, formal analysis, investigation. **Sijie Xian**: investigation. **Zhongyan Huang**: investigation, conceptualization, supervision. **Jingsheng Miao**: conceptualization, funding acquisition, writing – review and editing, supervision. **Chuluo Yang**: conceptualization, funding acquisition, writing – review and editing, supervision.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supporting File 1: anie72710-sup-0001-SuppMat.docx. The authors have cited additional references within the Supporting Information [50–52].

Supporting File 2: anie72710-sup-0002-cif.zip.